

Fórmula de la varianza

Proposición 1 (Fórmula de la varianza). Sea $X: \Omega \longrightarrow \mathbb{R} \in \mathcal{L}^1(\mathbb{P})$ v.a.

$$\implies \mathbb{V}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Demostración: Sea $X \in \mathcal{L}^1(\mathbb{P})$ una variable aleatoria, entonces

$$\begin{aligned} \mathbb{V}(X) &= \mathbb{E}[|X - \mathbb{E}[X]|^2] = \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &\stackrel{\text{lineal}}{=} \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \end{aligned}$$

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